

E scales with EH_x , where $E = C, O, N, S$ and $x = 1, 2, 3$, however, this scaling is not linear when comparing adsorbates that bind through different atoms (*e.g.* C versus O).

In Fig. S36 we show the correlation between the adsorption energies of EH_x , where $E = C, O, N, S$, and $x = 0, 1, 2, 3$. The structures (Fig. S36b) were relaxed with the MACE-MP-0 model with D3 correction (cutoff = 4 nm), and the adsorption energy was computed as $\Delta E_{\text{ads}} = E(a*) - E(*) - E(a)$, with a as the adsorbate and $*$ as the empty surface site. The observed correlations are linear in all cases except for the correlation between O and C. In Fig. S36, the MACE-MP-0 computed adsorption energies (blue/green/orange circles) are compared to the corresponding DFT values (connected with faint gray lines to faint gray circles) as reported by Nørskov (40). In this plot, although the absolute error of the obtained adsorption energies in comparison to the DFT values is high (which is not surprising as the model is extrapolating in this example), the trend of grouping metals into passive (noble, *e.g.* Au) catalytic (so-called Pt group) and non-reducible (*e.g.* Zr) is correctly captured and the essence of the LSR relationships was reproduced with the MACE-MP-0+D3 model.

A.24.3 CO (electro-)oxidation on Cu

We test the ability of MACE-MP-0 to predict the catalytic reaction mechanisms for the oxidation of CO on different facets of Cu, as previously explored via DFT (67). Specifically, we evaluate two complex potential-independent reaction steps of the CO oxidation, $\text{OH}^* + \text{CO}^* \rightleftharpoons \text{COOH}^*$ and $\text{OH}^* + \text{COOH}^* \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$ as shown in Fig. Fig. S37. Each reaction contains two reaction barriers, labeled i, ii and iii, iv, respectively. Other reaction steps of the CO oxidation which we do not include in our evaluation, are the adsorption of CO^* and OH^* , where the latter is an electrochemical process that can be described via the CHE (61). We recompute the reaction pathways for two Cu terraces (111) and (100) and two step-sites (110) and (211) via geometry optimizations of the initial and final states and subsequent NEB calculations (one for each reaction barrier i-iv) in lattice-parameter-adjusted simulation cells.

The reaction profiles and reaction barriers from the converged NEB calculations are shown in Fig. Fig. S37 for PBE+D3, MACE-MP-0+D3, and fine-tuned model (+D3). Fine-tuning was performed via a training set of 70 structures which was collected by drawing every fifth MACE-MP-0 NEB trajectory image and recomputing energies and forces with the computational settings of the MPtrj dataset. The foundation model MACE-MP-0+D3 shows in most cases qualitative agreement to the PBE+D3 reference, capturing the trends in barrier height differences for reaction steps i-iv. Quantitatively, the MACE-MP-0 barriers are underestimated, in line with systematic softening behavior noted in (265). Except to this are the cases of CO_2 and H_2O desorption (vi) for the (110) and (211) facet. Here, the barriers are overestimated due to an underestimated final binding energy and (relatedly) a different final state geometry. In contrast, the fine-tuned model shows near-quantitative agreement in terms of the barrier heights, deviating on average by only 0.056 eV. This close agreement does not become immediately apparent when comparing the NEB profiles which show some deviations. This can be attributed to a different number of images in the climbing image NEB calculations (43 for MACE and 19 for PBE+D3). To better demonstrate the accuracy of the fine-tuned MACE, we additionally performed PBE+D3 single point calculations for images of the fine-tuned NEB path, illustrating excellent agreement with the PBE-D3 energies.

A.24.4 In_2O_3

As a final test system we investigate a key step ($\text{CH}_2\text{O}_2 \rightarrow \text{CH}_2 + \text{O}$) in carbon dioxide hydrogenation to methanol over indium oxide via an NEB transition state search. This reaction has been extensively studied with *ab initio* methods due to indium oxide’s promising selectivity compared to conventional modified copper catalysts (41, 68). First, we perform a global geometry optimization of the reactant near an oxygen vacancy. MACE-MP-0 correctly identifies the three-oxygen-coordinated indium as the active site (41). Following a NEB calculation MACE-MP-0 predicts the reaction barrier within 15% of that investigated with DFT (1.16 eV vs. 0.98 eV), as visible in Fig. 2. Fine-tuning with just five single-point DFT calculation recovers the barrier with quantitative accuracy.

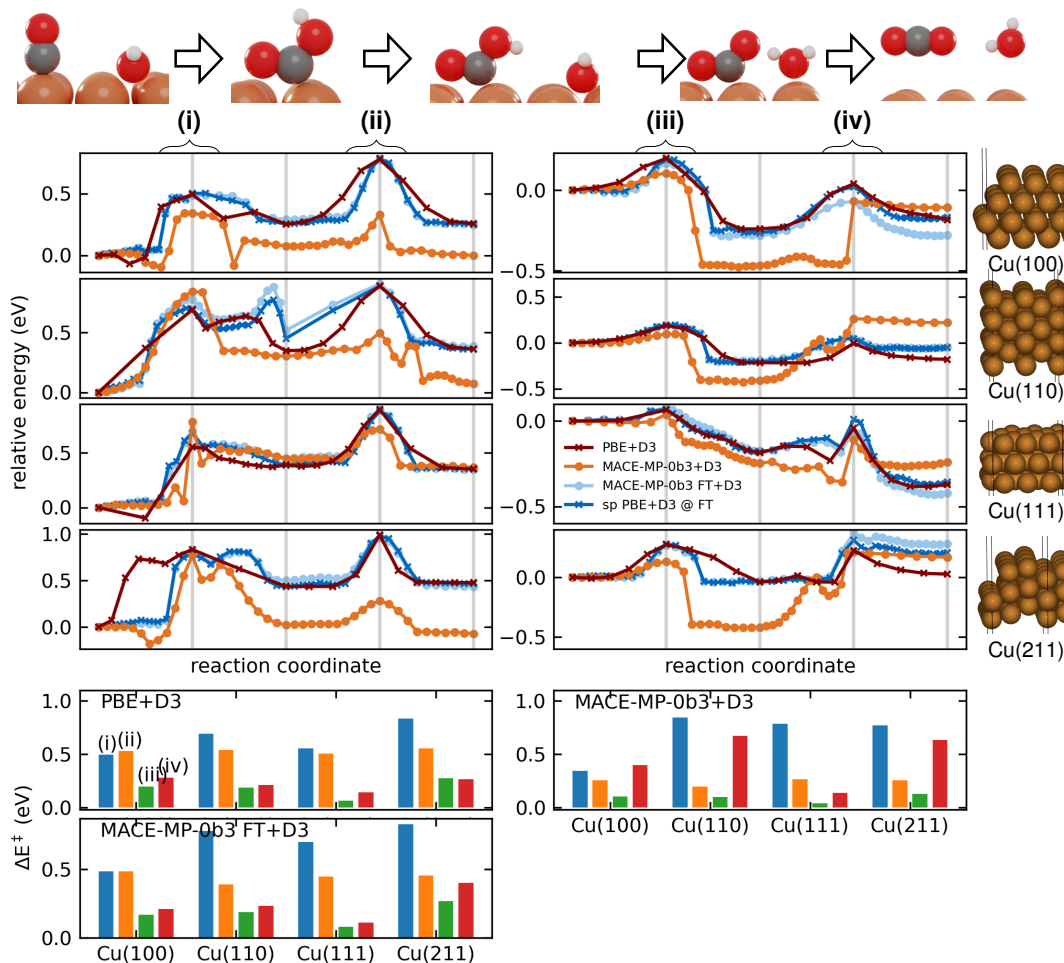


Figure S37: NEB profiles (top) and extracted barriers (bottom) of reactions (i-iv) computed for the potential independent steps in the multistep reaction mechanism of CO electro-oxidation on Cu for the low index facets (100), (110), (111), and (211). The reaction mechanism without implicit solvent is adapted from (67). The NEB calculations were carried out with MACE-MP-0b3 (orange), a finetuned model (lightblue) and DFT (darkred), each with the D3 dispersion correction. Additionally, single point PBE+D3 calculations (deepblue) were performed for the finetuned NEB-path. Faint vertical lines indicate minima and saddle points along the reaction coordinate for reaction steps i-iv.

Similarity statement

With the exception of the CuO bulk Pourbaix diagram, which is based on structures from the Materials Project, all presented examples treat surface slab models (with and without reacting adsorbates) and represent a significant extrapolation. The dataset does not include any such slab models but only related bulk structures. These bulk structures include 6 bulk structures that contain Pt, O, H and 15 bulk structures that contain Pt and O for the Pt Pourbaix diagram and the corresponding LSR example (similar number for other metals in the LSR), 8 different Cu-bulk phases and 111 structures composed of Cu, O, Cu, and H along with other elements for the example of CO oxidation on different Cu facets, and 9 structures that contain In, O, H and C and a 825 bulk structures that contain indium and oxygen for the In₂O₃ example. The most similar configurations for the CO oxidation example are Cu₂H₄C₄N₃O with the Materials Project ID mp-686268 and for the In₂O₃ example (NH₄)In(OH)PO₄ with the ID mp-764968. We provide Pt_LSR.json

and `COoxCu_closest_training_points.csv` for the LSR and CO oxidation example to help visualize the interactive UMAP on chemiscope.org.

Performance summary

Solid energies accurately predicted leading to correct Pourbaix diagrams. Adsorption energies overestimated, but linear scaling relationships between different surface/adsorbate pairs preserved. Minimum energy paths for reactive steps qualitatively correct, in some cases small (0.2 eV) in other larger (0.5 eV) energy errors.

A.25 Carborane rearrangement

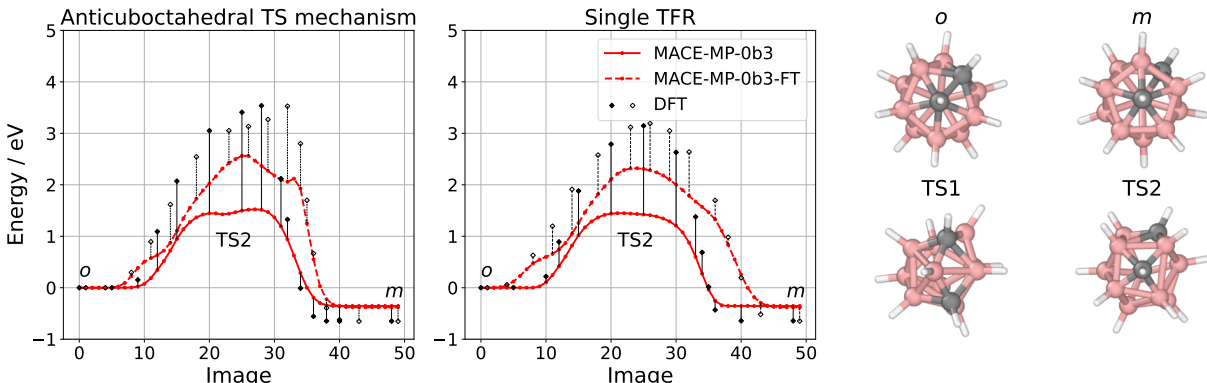


Figure S38: MACE-MP-0b3 and MACE-MP-0b3-FT NEB minimum energy paths for the isomerization reaction of *ortho*-carborane to *meta*-carborane. DFT energies, evaluated on selected structures, are shown. All energies are shown relative to the *ortho* isomer energy. On the right, end point and transition state structures are illustrated.

Carborane ($C_2H_{12}B_{10}$) is an organoboron compound with uses in drug discovery (266) and organometallic chemistry (267). It adopts icosahedral cluster structures, with three isomers based on different relative positions of carbon atoms: *ortho* (*o*), *meta* (*m*), and *para* (*p*). The thermally activated rearrangements between these isomers have been thoroughly studied (268), with several mechanisms proposed involving triangular face rotation (TFR). We used MACE-MP-0b3 to study two pathways from the *ortho* isomer to the *meta* isomer: one involving an anticuboctahedral transition state (by mutual rotation of two opposite faces), the other involving the rotation of a single triangular face.

We used MACE-MP-0b3 to obtain rearrangement pathways for each mechanism. The *o* and *m* isomer structures were relaxed with a force tolerance of $0.01 \text{ eV/\AA}$ using MACE-MP-0b3. Pathways were obtained using nudged elastic band with 50 images, relaxed with a force tolerance of $0.05 \text{ eV/\AA}$, first with spring constants of 0.5 and then $0.05 \text{ eV/\AA}^2$. We also evaluated energies of the images at the PBE/def2-TZVPPD level of theory using ORCA 5.0.3 (269).

Fine-tuning

We finetuned MACE-MP-0b3 using the *ortho*, *meta*, *para* structures and 7 high-energy structures from various pathways connecting these isomers. MACE-MP-0b3 and MACE-MP-0b3-FT pathways, along with some reference DFT energies, are shown in Figure S38. MACE-MP-0b3-FT gave larger energy barriers than MACE-MP-0b3: 2.56 eV vs 1.52 eV for the anticuboctahedral mechanism and 2.32 vs 1.45 eV for the single TFR mechanism. While fine-tuning improved energy errors in the high-energy regions, the reaction energy errors were not improved significantly. The reference DFT reaction energy is -0.647 eV while MACE-MP-0b3 and MACE-MP-0b3-FT predicted -0.355 and -0.380 eV, respectively.

Similarity statement

The training set contains 5 structures composed of H, B, C elements, and 21837 structures that have H or B or C along with any other elements. The dataset contains 2 structures or containing icosahedral $C_2H_{11}B_{10}$ clusters linked by a C-C bond to form a dimer, 1 icosahedral borane $B_{12}H_{12}$ cluster and 18 other borane clusters containing trigonal B_3 faces. Along these, the dataset contains several hundred structures involving derivatives of borane and carborane clusters such as salts, metal complexes, and halogenated species. The closest structures in the training set are mp-1194548 ($C_4H_{22}B_{20}$ containing two icosahedral carborane clusters joined with a C-C bond), and metal complexes containing carborane ligands: mp-759303, 705569, 1199795, 1198024.

Performance summary

Carborane rearrangement pathways found with MACE-MP-0b3 are qualitatively correct but do not lie on the reference DFT minimum energy path, as shown by the >1 eV energy errors relative to DFT in the high energy regions of the pathway.